

Nutrients status in Kundu River, part of Kadapa dist, A.P: Implications for environmental proxies.

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Abstract:

Investigation of agriculture, industrial waste had resulted in increased soil degradation and erosion, with associated pollution of surface waters. Small field wetlands, constructed along runoff pathways, offer one option for slowing down and storing runoff in order to allow more time for sedimentation to be taken up by plants or micro-organisms. The representative soil profile in Kundu river in kadapa district was selected for the study. At location, composite soil and water samples were collected from different horizons viz., 0-5 and 10-12 cm segments. The soil texture usually, ranged from sandy, clay and loam. Amount of sand, silt and clay in the sediment profile varied from 32.6-49.2- 32.6, 20.1-28.7 and 27.6- 34.2% respectively. The water samples (3in number) were collected from different places of Kundu river. The quality analysis has been assessed through the pH, EC, TDS, Dissolved Oxygen, BOD, COD, Total Hardness, Sodium, Potassium, Calcium, Magnesium, Chloride, Fluoride and Alkanity and heavy metals. Heavy metal levels in water and sediments were analyzed by Inductively Coupled Plasma Spectroscopy (ICP/OES). The obtained results showed that the average values of ferrous sulphate in water samples were higher than the respective reference values for fresh water. The analyses of heavy metals (Fe, Cd, Cr and Pb) in water and sediment were carried out. Heavy metals present in the environment possess significant global problem and warrant ultimate remediation measurement. In the present study, bacteria were isolated from the Kundu River, kadapa. The three isolates were further selected on the basis of tolerance to various heavy metal concentration. Minimum inhibitory concentration

test recorded higher level of tolerance (5 - 10 mM) to the tested. The one heavy-metal resistant bacteria of present study could survive well in the heavy metal contaminated environment. Therefore, it can be exploited for the remediation of metals from such environment.

Key words: Nutrients status, Heavy metals, Physico chemical parameters, Kundu river.

Introduction

At the present, rivers, and next to them, bays, lakes and oceans are most polluted water bodies in that order and about 80% of diseases are attributed to polluted water (Chakraborty et al. 2021). River is one of the most important natural resources. However, an increasing population infected especially by heavy metal pollution through which water can enter the body and cause various diseases (Khatun et al. 2021). Rivers play a major role in integrating and organizing the landscape, and molding the ecological setting of a Basin (Chauhan et al. 2020). They are the prime factors in controlling the global water cycle and in the hydrologic cycle (Skiris et al. 2020). Rivers are the most dynamic agents of transport. River water has been in use for several purposes such as, drinking, industrial, domestic, and agricultural needs (Singh et al. 2020).

Soil is the most important basin nutrient resource that determines the ultimate sustainability of any agricultural system (Stephens et al. 2020). The available macro and micro nutrients in the area help to determine the soil potential to supply nutrient for crop growth (Johnson et al. 2021). The functioning of an ecosystem and its stability to support life forms depend, to a great extent, on the quality of its water (Hamid et al. 2020). The key feature of an ecosystem is the interaction among the biotic and abiotic components (Cruz-Paredes et al. 2021). Any change in the abiotic components will be reflected in the biotic (Janecke et al. 2020). The modern civilization; Industrialisation, urbanisation and increase in population have lead to the degradation of our water quality (Khan et al. 2021). Quality of

water generally refers to the component of water, which is to be present at the optimum level for suitable growth of fauna and flora (Arif et al. 2021). Various factors like temperature, turbidity, nutrients, hardness, alkalinity, dissolved oxygen play an important role for the growth of plants and animals in the water body (Sarkar et al. 2020). However, on the other hand, biological oxygen demand, chemical oxygen demand indicate the pollution level of the water body.

Obviously, the surface water quality is deteriorating due to anthropogenic activities, industrialization, farming activities, transportation, urbanization, animal and human excretions and domestic wastes. Variation in the quality and quantity of river water due to natural and anthropogenic activities has been widely studied with regard to several rivers, world over.

Study area

The Kundu River is a tributary of the Penna River in the Rayalaseema region of Andhra Pradesh, India. Originating as a spring near the village of Uppalapadu in Orvakal Mandal of Kurnool District, it goes through many changes before merging the Penna at Adinimmayapalli village of Kadapa district (Fig. 1). It is known for frequent floods that bring heavy damage to the Nandyal and Koilkuntla areas, and hence it is popularly called the "Sorrow of Nandyal". But now a days Nandyal town drainage system has been attached to Kundu River without filtering the drains. Industrialists focus their vision more and more at the profits and cause pollution to the river as maximum as possible. The villagers who are living by the banks of river develop skin diseases. The pollution impacts even the health of animals too. In ancient times this river was known as the Kumudvathi. There is a saying in Rayalaseema that whoever drinks the water of the Kundu will gain enormous courage to face enemies.

The Kundu River irrigates about 5,000 hectares in its valley, consisting about 50 villages. It covers 6 assembly constituencies in Kurnool and Kadapa districts. Many streams and brooks, including the Gaaleru, Paller, Nippuvaagu and Sankalavaagu, are the main sources of flood water. Gaaleru and Paaleru in particular bring heavy floods from the Nallamala hills. The Kundu is ferocious in the rainy season, particularly during cyclones.

2. Materials and Methods

2.1 Study area

River sediment and water samples were taken from three areas in Kundu River lies between latitude 14 53 07 N- 14 56 09 N & 78 55 10 E- 78 59 06 E (Fig. 1).

2.2 Sample collection:

Site 1

The first sample collection area is at the depth of 0-5 cm interface of sewage with Kundu River entrance, in Chappadu (village).

Site 2

The second sample collection area is at the depth of 5-10 cm of agricultural land of with Kundu River in near kethavaram village.

Site 3

The third sample collection area is the depth of 10-15 cm site of interaction of the agricultural land with Khagipet village.

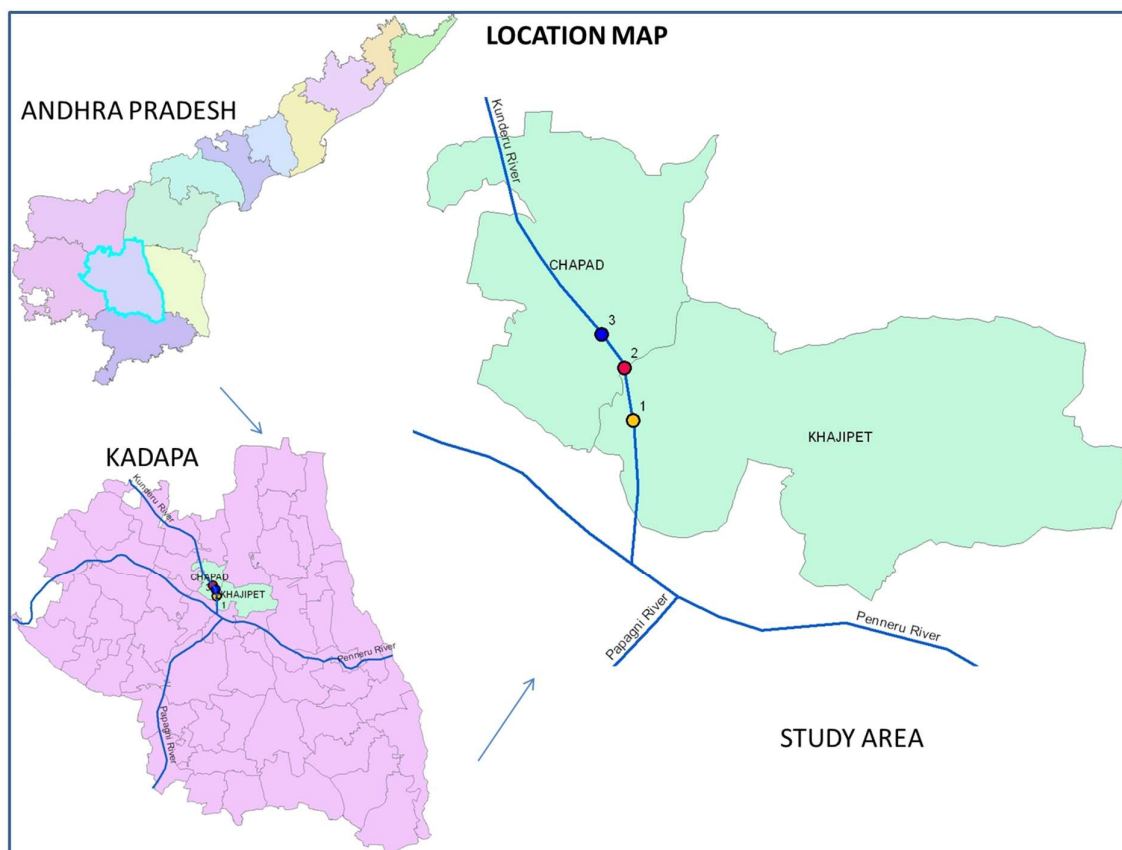


Fig: 1 Location Map of the study area

2.3. Analysis of physico-chemical parameters

The collected water samples were analysed for pH, temperature, electrical conductivity, alkalinity, TDS, Chloride, nitrate, Acidity, COD, Calcium hardness, Salinity, BOD, Fluorides, Dissolved oxygen (DO), metal concentrations, specific anion concentrations, and nutrients availability (Mekapogu et al. 2018) (Table. 1).

2.4. Heavy metal analysis by ICP-OES

Inductively Coupled Plasma Atomic Emission Spectroscopy (ICPAES), also referred as inductively coupled plasma optical emission spectrometry (ICPOES), is an analytical facility used for the detection of trace metals (Mekapogu et al. 2018). It is a type of emission spectroscopy that uses the inductively coupled plasma to produce excited atoms and ions that emit electromagnetic radiation at wavelengths characteristic of a particular element. The

intensity of this emission is indicative of the concentration of the element within the sample.

The values are given in Table. 2.

2.5. Nutrient status in sediment of kundu river:

Mineral matter of soil samples such as sand, silt, and clay contents were analyzed by means of different sizes of sieves by following the method described by Alexander (1961). The available of the nutrients as nitrogen was determined by alkaline potassium permanganate method. And available phosphorus was extracted by Olsens method. Available potassium and sulphur was determined by Flame photometer.

2.6. Culture media condition:

Nutrient broth was prepared by adding peptone – 0.5 %, yeast extract- 0.3 %, NaCl – 0.5 % , Glucose - 0.25 % . The pH was maintained to 7 at room temperature. Then nutrient agar plates were required approximately 1.5 % agar was added to nutrient broth (Sirisha et al. 2017).

2.6.1 Isolation of heavy metal resistant bacteria:

For the selective screening of heavy metal resistant bacteria, 300 µg/mL of heavy metal (Lead) incorporated LB (Luria Bertani) agar plates (Peptone 10.00 g/L, yeast extract, 5.00 g/L, NaCl 5.00 g/L, dextrose anhydrate 10.00 g/L and agar 30.00 g/L: pH -7.00) were used and screened by standard serial dilution method observed at 37 °C. The plates were observed after 24 hours of incubation for any kind of development on the culture medium. After preliminary screening of samples containing heavy metal degrading isolates, serial dilution was done to isolate desired bacteria Azad et al. (2013). Streak plate technique was followed during isolation. Control plates also prepared with LB media without including any heavy metal to make further comparison. Colonies differing in morphological characteristics were selected, picked, purified and then preserved on different plates for further studies (Sirisha et al. 2017).

2.6.2 Biochemical tests

Biochemical analysis was done in order to know which type of bacteria are present in the water samples of the study. The biochemical tests include catalase, Indole, Citrate, Urease. These tests were performed according to the standard methods (Sirisha et al.2017).

2.6.3 Minimum Inhibitory Concentration (MIC):

The MIC of heavy metals Cadmium (Cd) for selected bacteria isolate was determined by the plate dilution method as adopted (Aleem et al. 2003). The heavy metals were used in different concentrations varying from 10 to 50 mM. The growth was observed after 24 - 48 h at 37°C. There were three replicates for each concentration of heavy metal used. The lowest concentration, at which bacterial growth prevented, was considered as MIC.

2.6.4. Antibiotic sensitivity Test:

Antibiotic sensitivity was done by cup assay method. Overnight culture of the tested bacteria was spread on nutrient agar media plates. These plates were incubated at 37°C for 24 hours. Each antibiotic at their concentration of 30 mcg Amoxillin, Cefolac, Ciprofloxacin, Cefadroxil, Gentamycin, Cifran were poured on cups of pre inoculated plates. All plates were incubated at 37°C for 48 hours. The inhibition one around each cup was observed against each antibiotic and also against each bacterium.

2.6.5. Effect of Iron (Fe) metal on Bacterial growth:

Nutrient Broth at 37°C /24h and 10 µL of cultures were placed on Nutrient Agar with different concentrations of the respective metal (1-4mM). The same media without metal was used as control.

3. Results and Discussion

3.1 Analysis of physico-chemical parameters:

The Physico-chemical properties of river were provided (Table 1). The pH range in the study was between 6.53 to 6.90 indicating the water is near to neutrality in nature. The

variation can be due to the exposure of river water to atmosphere, biological activities, and temperature changes. The temperature range between 35°C to 37°C was observed. The EC values represent the soluble salts concentration in water, and its higher content may cause gastrointestinal irritation as the high EC represents the elevated salt content on water. The EC ranges between 1.20 to 1.56 mS. The sample 3 has registered high value 1.56 mS among others. A significant correlation was observed among EC values and hardness, calcium, and chloride ions. EC is one of the important factors influencing crop yield. As the EC of water goes up, plants need to compete more with free ions in water for intake of water for plant growth. This means that an increase in EC reduces the availability of water to the plants (Prasad et al. 2016)

Alkalinity is a measure of the ability of water to neutralize acids. The relative quantities of the alkalinity species are pH dependent. The Alkalinity values in the present study region ranges between 24 - 36 mg/L. This indicates that bicarbonates form the dominant anion in the system. This may be due to the phytoplankton growth. The growth will be usually higher if organic wastes are being added to the water. Except one sample, all the sample values were above the desirable range. Among the 10 samples, three samples – Sample 2, 5, and 8 have recorded carbonate alkalinity in the range of 6-32 mg L⁻¹ (Prasad et al. 2016). And the TDS range of this present study is between 0.89 to 0.93 ppm 919. Similar results were reported by Mekapogu et al. 2018, in river thungabadhra which showed the TDS value is 0.92 to 1.34 in three sites.

The range of Chloride was 0.03898- 0.03998 mg L⁻¹, and the values indicate that they are in the desirable limits. Namrata 2010 reported that Chloride is one of the major inorganic anion in water and waste water. The chloride written shows variation with a range of 8.2 (July) at Shiwala Ghat to 81.5mg/L (Feb) at Raj Ghat. The Chloride vales remain higher in rainy season in river ganga at Varanasi.

The range of nitrate was observed between 1.03 ± 0.05 to 5.7 ± 0.07 . Site 3 showed the highest concentration of nitrate. The range of acidity was observed in three sites between 49mg/l to 59mg/l. The highest acidity was observed at site 2 sample.

The levels of Chemical Oxygen Demand (COD) recorded in three sites is 147.2 mg/l, 145.8 mg/l, 147.5 mg/l respectively. The COD levels in our study exceeded the WHO standards 50 mg/l (WHO, 1990), while investigating pollutants, determining physico-chemical characteristics of the Nairobi River and remediation of some toxic heavy metals using Fish Bones' across the sampling sites recorded COD with a mean of 260.6 ± 248.8 mg/L. There was a significant difference in COD ($t=4.570$; $df=11$, $p=0.001$) between Nyaguru (upstream) and Daraja (Downstream) of Nyakomisoro spring. There was a general decrease of COD values downstream indicating that amount of organic matter was decreasing gradually towards downstream (Kerich & Fidelis, 2020).

Biochemical Oxygen demand (BOD) is the quantity of dissolved oxygen (DO) needed by aerobic biological organisms in a body of water to break down organic material present in a given water sample at certain temperature over a specific time period (Masese, 2010, Walakira & Okot-Okumu, 2011). The BOD levels at sites 1, 2, & 3 are 14.1 mg/l, 14.7 mg/l, 8.9 mg/l respectively. Water hardness arises due to the presence of cations such as calcium and magnesium and anions such as bicarbonates, chlorides, and sulfides [Rousseau et al. 2008]. In the present study, the range of hardness is 25.23 mg/l to 50.463 mg/l. Among the samples sites, all the samples have registered lower value than the WHO limit. Calcium ions usually increase the hardness content in the river water. The range of Ca^{+2} ions is 119.7-212.66 mg L^{-1} ; the higher values and the strong relationship with EC, hardness, alkalinity indicates that calcium forms the dominant cation in the Penna river and readily associates with bicarbonate system of fresh water bodies. Except sample 3, all the nine samples figured below the WHO desirable limit (Bureau of Indian Standards 1998 & 2012, WHO, 1996).

The Salinity of three samples were observed in the range of 0.02 to 1.09 ppt. Fluoride concentration varies in site 1, 2 and 3. In the present study region it ranges between 1.33 – 3.70 mg/L. The desirable limit in drinking water is between 0.5 to 1.5 mg/L. The permissible limit in drinking water is about 1.5 mg/L [WHO 1990]. The higher concentration of fluoride observed in site – 2 (3.70 mg/L) and site – 3 (2.10 mg/l) exceeding the permissible limit of W.H.O limit (1.5 mg/L). This may be due to anthropogenic activities including the lithological constitution of the region. The measured Dissolved oxygen levels in sites 1, 2, & 3 are 4.0 mg/l, 3.8 mg/l, 4.2 mg/l respectively. Kerich & Fidelis (2020) reported that the Low levels of DO in River Nyakomisaro were recorded. This could be attributed to increase flow rate which increases turbulence causing aeration. In addition, the low dissolved oxygen may be as a result of decomposition of submerged vegetation which was not deforested during construction of the river. Recorded in C1 the levels in mg/l ranged from 5.3 to 7.34; 5.7 to 6.4 in C2; 1.7 to 6.46 in C3 and 2.2 to 6.9 in C4.

Table.1 Physico-chemical parameters of kundhu river water

Property/Site	SAMPLE 1	SAMPLE 2	SAMPLE 3
Depth(Cm)	5 cm	5cm	10 cm
Temperature (°C)	36	37	35
p ^H	6.53	6.70	6.90
Conductivity (mS)	1.37	1.20	1.56
Alkalinity	24mg/l	36mg/l	24mg/l
TDS	0.89ppm	0.90ppm	0.93ppm
Chloride	0.03898	0.03998	0.03998
Nitrate	1.06±0.02	1.03±0.05	5.7±0.07

Acidity	49mg/l	59mg/l	53mg/l
COD	147.2mg/l	145.8mg/l	147.5mg/l
Calcium hardness	25.23mg/l	19.8245mg/l	50.463mg/l
Salinity	0.02 ppt	1.09ppt	0.02ppt
BOD	14.1	14.7	8.9
Fluorides	1.33mg/l	3.70 mg/l	2.10mg/l
Dissolved oxygen	4.0mg	3.8mg/l	4.2mg/l

3.2. Heavy metal analysis of Kundu river water by ICP-OES:

Assayed Fe showed high metal concentrations in sediments of all three sites (Table. 2).

Table 2. Heavy metal analysis of Kundu river water by ICP-OES

Metals	Cd	Cr	Fe
Site 1	0.467ppm	0.039ppm	2.152ppm
Site 2	0.427ppm	0.023ppm	2.102ppm
Site 3	0.398ppm	0.030ppm	2.159ppm

3.3. Nutrient status in sediment profile:

It is a nitrogen status of soil profile ranged between 218.5-230.2 kg ha⁻¹. The maximum amount of available nitrogen was found in the surface. The phosphorus status of soil profile ranged between 11.1-14.8 kg ha⁻¹ with a mean value of 12.5 kg ha⁻¹. The maximum amount of available phosphorus was found in the surface. The potassium available in the soil profile 175.8- 198.6 kg ha⁻¹. Maximum amount of available potassium was found in the surface. The low amount of sulphur available in the soil profile 8.2 Kg ha⁻¹ (Table .3)

Table 3. Nutrient status in sediment profile

S.NO	Depth cm	Sand%	Silt%	Clay%	Avaliable nutrients Kg ha ⁻¹			
					N	P	K	S
1	0-5	49.2	20.2	27.6	230.2	14.8	198.6	8.2
2	5-10	42.6	21.8	30.8	218.5	12.5	195.5	9.5
3	10-15	40.2	22.3	32.3	220.5	11.1	175.8	6.2

3.4.Isolation of heavy metal resistant bacteria:

Serial dilution method and spread plate method were used to isolate the heavy metal resistant bacteria. In the present study, a total three colonies were isolated and they were maintained on nutrient agar plates, stored at 4 °C for regular use. All three were designated as KR 1, KR 2, KR 3.

3.4.1. Morphological characterization of heavy metal resistant bacteria isolated from Kundu river water sample:

In the present study, the morphological characterization such as color, size, shape and nature of colony of all three were determined by growing them on nutrient agar plates. In addition, all the isolates were tested by grams staining to check whether they belong Gram positive and Gram negative. The details of the morphological studies were shown in table 2. Most of the colonies were isolated during this study were irregular in nature. The colour was generally Yellow, white and creamy and their shape is rod to circular. Out of three isolates isolated from Kundu River water two strains were Gram positive rods and cocci and another one was Gram – negative cocci forms. The bacteria isolate KR1 appeared as Yellow gram – positive rod shaped bacteria. KR2 and KR3 were gram negative and Gram positive bacteria appeared as White and creamy color which were in cocci forms.

Table 4: Morphological characterization of isolates

Strains	Cell Morphology			Gram's Test
	Shape	colour	Margin	
KR1	Rods	yellow	Entire	+ve
KR2	Round	white	Entire	-ve
KR 3	Cocci	creamy	Entire	+ve

3.4.2. Biochemical characterization of bacteria isolated from Kundu river

After performing the following tests, it was deciphered that the strains organisms 1, 2, showed negative test for indole and organism 1&3 are positive for Citrate and catalase and respectively organism 2 was negative. Organism 1& 3 are negative and organism 2 was positive for urease test (Table. 5).

Table 5: Biochemical test for isolated KR 1, KR 2 & KR 3

Sources	KR1	KR2	KR3
Catalase	+ve	-ve	+ve
Indole	-ve	-ve	-ve
Citrate	+ve	-ve	+ve
Urease	-ve	+ve	-ve

3.4.3. Antibiotic Sensitivity test for isolated KR 1, KR 2 & KR 3:

All three isolates (KR 1, KR 2, KR 3) have been checked for antibiotic sensitivity. All most all isolates have shown sensitivity to some antibiotics and resistant to some other antibiotics but the isolates KR 3 has shown resistant to many antibiotics compare to other isolates. The response of the other isolates to different antibiotics were shown in table 6.

Table 6: Antibiotic Sensitivity test for isolated KR 1, KR 2 & KR 3:

S.no	Antibiotics	INHIBITION ZONE(cm)		
		KR1	KR2	KR3
1.	Amoxillin (5µg)	2	2.2	1.8
2.	Cefolac (30 µg)	1.9	1.8	1.2
3.	Ciprofloxacin(20 µg)	3.7	3.5	2.8
4.	Cefadroxil (15 µg)	2.5	2.4	2.6
5.	Gentamycin(10 µg)	3.3	4.2	2.8
6.	Cifran(20 µg)	3.0	2.7	3.2

3.4.4. Effect of Iron (Fe) on bacterial growth:

All isolates have been shown growth was decreased with the increasing concentrations of metal (Fig. 2).

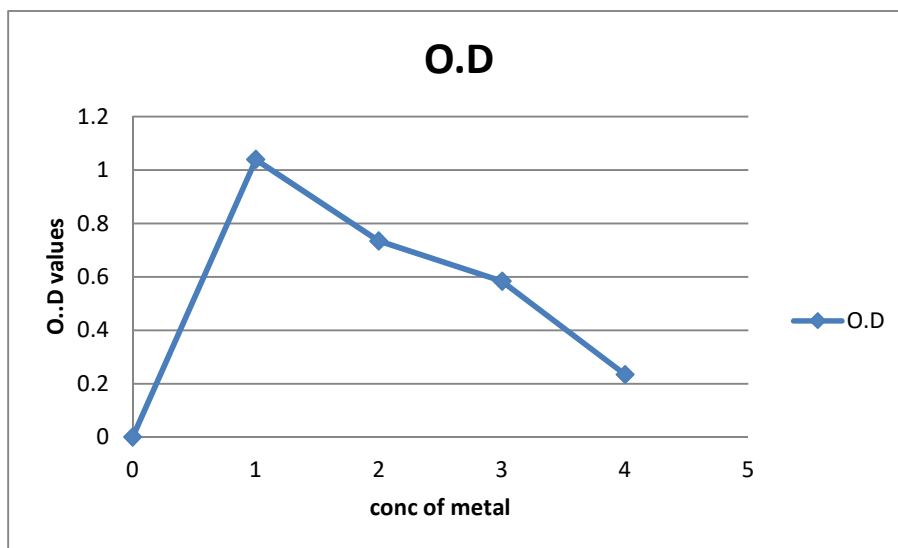


Fig 2. Effect of Iron (Fe) on bacterial growth.

Conclusion:

The present work aims to know wellness of the river body by ascertaining the physicochemical parameters for water and nutrient status of the sediment samples. In addition the present work seeks to characterize and inmate relation of heavy metals with microbes. Most of the physico-chemical parameters in water were present at higher levels than the WHO permissible limits. This study clearly observed that the river area is highly contaminated by pathogenic bacteria that too mainly with gram positive and less with negative rods. If this water is used for domestic purposes without proper treatment it may lead to serve illness to human and animals too. This will affect mainly children living in that particular geographical area. Therefore, it recommends the need for establishing a proper treatment plants to remove all contaminated bacteria before they will be utilized for domestic purposes.

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